# RealQM Molecule Simulator: Validation on Small Molecules from Hydrogen and the Second Row

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*Mathematical formulation by the author; implementation and numerical experiments developed with AI assistance (Claude, Anthropic).*

**Abstract**

We describe a simulator for small molecules built from hydrogen, carbon, nitrogen and oxygen in RealQM, a real-space model in which each electron is a unit-charge density occupying a non-overlapping spatial domain, and the validation it is built to support. The model is given three valence charges—$Z_x = 1, 3, 4$ for hydrogen, carbon, and nitrogen with oxygen—and each molecule's observed geometry; it returns the binding energy and the geometry it settles at. Every other quantity is a solver setting, and the settings are declared once and never adjusted per molecule. That is a weaker claim than parameter-free and a more useful one: the construction has settings, which we tabulate, but no per-system adjustment, so each molecular number is a prediction rather than a fit. The central such declaration is the grid: one spacing, $h = 0.1\,a_0$, for every molecule in the list, treated as a commitment about what the model is rather than as a convergence parameter. At fixed $h$ the grid size no longer carries resolution—it fixes the box—and the box must be fixed too, because mixed boxes were measured to corrupt a cross-molecule energy difference by a factor of 2.7. The declaration costs something and buys something. It costs the claim of being parameter-free: $h$ becomes a constant of the model, fixed by measurement rather than derived, and we say so plainly rather than presenting the grid as a numerical setting. It buys falsifiability, since with $h$ free any disagreement can be deferred to convergence, whereas with $h$ declared every bond length and every binding energy is a prediction carrying no remaining adjustment. On that footing the simulator reports both observables directly: binding energy against the measured well depth $D_e$, and bond length against $R_e$. We work hydrogen through in full as the foundational case. Its binding energy is $0.1666 \pm 0.0018\,E_h$, where the uncertainty is measured scatter over a factor of two in the initialisation and not an estimate. That value falls *between* the two experimental conventions—4.5% below the well depth $D_e$ and 1.2% above the directly measured $D_0$—which differ by more than the model's own scatter, so the model is accurate to a few percent on either reading and the sign of its error is set by a convention it cannot resolve. The declared box is worth a further four points, more than the scatter, and runs the unflattering way: a smaller box binds more strongly, so enlarging it costs accuracy rather than buying it. Its bond is 52% too long; and these are not independent defects but one, since the well's curvature is 49% too soft and $\Delta R = F/k_{\text{model}}$ turns a given force residual into a proportionately larger displacement—a conversion that must use the model's own curvature, the measured force constant understating the bond error by the stiffness ratio. Its bond error is almost entirely discretisation rather than model: an axisymmetric reduction, exact for a two-domain system on an axis, converges the same bond to $1.368\,a_0$—short by 2.3% and flat to 0.18% under a 2.5-fold refinement—with the energy minimum and the force zero agreeing to 0.06%. So the refinement sequence crosses experiment on its way to

something shorter, and the spacing at which the error vanishes marks a cancellation rather than an agreement. The two quantities we validate on are the *geometry the model settles at* and the *binding energy*—a length and a depth, both in the units the model claims to predict. Forces at observed geometries, which an earlier version of this work made primary, are demoted to a check: a residual converts to a bond error only through the model's own curvature, and using the measured force constant instead—as we did—understates hydrogen's error by twenty-five points. What the force retains is narrower and real: the total residual measures electronic non-convergence, and its agreement with $-dE/dR$ (0.06% in an exact axisymmetric reduction) is what licenses using forces for dynamics. It is not a figure of merit, and an unbound hydrogen molecule scores as well on it as the best entries in the list.

# 1 The simulator

The simulator presents a list of molecules, each stored as experimental internal coordinates converted to Cartesian form and centred on the centroid. Selecting one builds the system on the declared grid of Section 4—$h = 0.1\,a_0$, $200^3$, box $20\,a_0$, identical for every entry—and runs. The solver is a GPU code (`mol_fast`) carrying one domain per entry and capped at sixteen domains; a second implementation of the same model as one globally partitioned field admits two thousand and is discussed in Section 7.

Four choices are built in as defaults rather than offered as options, each because it was measured rather than assumed. The grid is fixed, for the reasons of Section 4, and does not follow the molecule—an earlier version sized the box from each molecule's own extent, which is what makes a table of cross-molecule differences unusable. Valence charges are $Z_x = 1, 4, 3, 3$ for H, C, N, O—oxygen at +3 rather than the common +2, which converges five times worse at the same bond error, and carbon never at +2, which has no defence (Section 13). The nuclear force is the volume integral of Section 3.1 rather than a sampled potential gradient. And the kinetic energy of partially occupied interface cells is excluded above an occupancy of 0.9, which is the single change that brings hydrogen's binding energy from $-7.6\%$ to within its own scatter of $D_e$ (Section 6).

The kernels are held fixed by default, since the measured geometry is the input the whole method rests on. They can also be released to move under the force, which is the sharpest available test of the force and is reported for $H_2$ in Section 14. It is offered with the restriction measured there attached: relaxation is trustworthy for geometry but not for energies, and at three centres the electronic state does not settle, so released kernels are at present a two-centre capability rather than a feature of the molecule list as a whole.

The two observables that can be compared with experiment are reported directly in the output. The bond lengths and angles are displayed against their measured values, and the binding energy is displayed as

$$E_{\mathrm{bind}} \;=\; \sum_i E(\mathrm{atom}_i) \;-\; E(\mathrm{molecule}), \tag{1}$$

positive for a bound molecule, against the measured well depth $D_e$ where a verified value exists. The atomic references in (1) are stored *keyed by grid*, and a run whose grid has no reference set reports no binding energy together with the reason, rather than silently subtracting an energy computed on a different mesh. This is the declaration of Section 4 doing its work: once the grid is fixed, the atomic energies are constants of the model rather than per-run bookkeeping.

What further distinguishes the simulator from a calculator is that it displays its own error checks continuously alongside the result: the kernel–kernel energy against exact arithmetic on the geometry actually built, and the total nuclear force as a percentage of the largest single force. The reasoning behind those two quantities, and why which of them carries information depends on the

molecule's symmetry, is the subject of Section 9, where we also explain why the second is useful as a convergence diagnostic but must not be read as a figure of merit. Their practical effect is that a calculation which cannot be trusted says so on the same screen as its answer.

## 2 Introduction

RealQM [1] replaces the many-electron wavefunction by $N$ unit charge densities occupying non-overlapping spatial domains with free interfaces, interacting by Coulomb's law with each pair counted once. Nuclei enter as kernels; in molecular work a filled core is replaced by a *pseudo-kernel* of reduced charge $Z_x$ carrying only the bonding valence. The model is therefore a force law: kernels move under the electrostatic force of the charge distribution, and a predicted geometry is wherever that force vanishes.

**What goes in, and what comes out.** The model is given *three valence charges*— $Z_x = 1, 3, 4$, covering hydrogen, carbon, and nitrogen with oxygen—and each molecule's *observed geometry*. Everything else is a solver setting, and the settings are declared once and never adjusted per molecule. That is the load-bearing claim of this paper, and it is weaker than "parameter-free" while being more useful: the construction plainly has settings, listed in Section 4, but it has no *per-system* adjustment, so every molecular number is a prediction rather than a fit. A reader can check this by inspection, which is why the settings are tabulated rather than described.

What comes out is the *binding energy* and the *geometry the model itself settles at*, with the *dipole* as a third quantity that doubles as an acceptance test—a symmetric molecule must give $\mu \equiv 0$, so any value there is pure error and rejects the state before its energy is read.

Two limits of that economy belong here rather than in a closing section. Three charges covering four elements is *degenerate*: nitrogen and oxygen both take $Z_x = 3$ and are bit-identical calculations, so the model cannot distinguish them and no result involving both speaks to their difference. And the geometry prediction presently covers four molecules rather than thirteen, because rescaling a bond is implemented for diatomics only and three-centre relaxations oscillate (Section 16).

Validating such a model is awkward by the usual routes. Scanning the energy over geometry requires every point of the scan to be converged, and a scan assembled from partially relaxed states returns a smooth but meaningless curve. Relaxing under the force requires the force to be trustworthy, which is the thing under test. Both routes also need many calculations per molecule, and both return an answer whether or not the underlying states deserve one.

**Observable and computable.** A quantity is *observable* if an experiment on the real system yields it, and *computable* if this model yields it. The two are independent, and sorting the candidates by both at once decides what can be validated and what cannot:

| | **computable** | **not computable** |
|---|---|---|
| **observable** | dipole moment; geometry | $D_0$ |
| **not observable** | $D_e$; $E$, $T$, $V_{eK}$, $V_{ee}$; $\sum_a \mathbf{F}_a$ | — |

Only the upper-left cell can test a theory, and the dipole is the one output that sits there without qualification: directly computed here and directly measured in the laboratory, with no inference in between.

The geometry needs a qualification, and it is the same one the energy needs. Having no nuclear motion, this model finds the minimum of its own curve—an equilibrium quantity. The matched

experimental partners are therefore the *equilibrium* ones, and both are reached from measurement by extrapolation rather than read off:

| | well minimum (what the model gives) | vibrational ground state |
|---|---|---|
| length | $R_e = 1.40028\,a_0$ for $H_2$ | $R_0$, an average over $v = 0$ |
| energy | $D_e = 0.17447\,E_h$ | $D_0 = 0.16456\,E_h$ |

$R_e$ comes from rotational constants $B_v$, one per vibrational level, each giving a $\langle 1/r^2 \rangle$ average, extrapolated to the bottom of the well through a vibrational analysis; $D_e$ comes from $D_0$ plus a zero-point correction from $\omega_e$. So *both* of this paper's primary comparisons are with quantities that spectroscopy infers rather than records, and its geometric *input* is inferred the same way. That is worth stating plainly: this is not experiment in and experiment out, and the honest claim is that the model predicts the equilibrium quantities spectroscopy infers—with the dipole as the single exception needing no inference.

The other cells explain two difficulties that would otherwise look like bad luck. A total energy is computable and not observable: only differences are measurable, so the $-1.174470\,E_h$ this work compares against is Kolos–Wolniewicz's *calculation* in the Born–Oppenheimer limit, and agreeing with it is agreeing with a computation. And the dissociation energy is split across the diagonal: $D_0$ **is observable but not computable here**, because the model has no nuclear motion, while $D_e$ **is computable but not observable**, no experiment measuring a well depth directly. The six-point gap between them is exactly the zero-point energy—the quantity the model omits. So quoting both is not a hedge between conventions; it is the only honest response to a comparison that must cross between cells.

$\sum_a \mathbf{F}_a$ is in the lower-left cell, which is the deepest reason it was never a figure of merit—deeper than the symmetry argument of Section 9. It is not a physical quantity at all.

We therefore validate on two quantities, with a third kept as a check rather than a result.

**The geometry the model settles at—primary.** The separation and angles the model itself reaches, against the measured ones. This is the observable with the fewest defects: it is a *length*, in the units the model claims to predict; it needs no second quantity to interpret; and it is not symmetry-protected, where a force residual is—$CO_2$ can contract symmetrically with $\sum_a \mathbf{F}_a$ staying exactly zero while every bond is wrong. It must be taken from a *commensurate, two-sided* scan: released from the experimental geometry the molecule expands and stops short, giving $1.796\,a_0$ for $H_2$ against a true minimum of 2.134, so one-sided relaxation flatters the model by some ten points.

**The binding energy—primary.** $E_{\text{bind}}$ from (1) against the measured well depth, reported against both conventions because they differ by more than the model's own scatter. It requires atomic references computed with the same construction on the same grid, which is why Section 5 treats the atom before any molecule.

**The force at the observed geometry—a check, not a figure of merit.** At the measured equilibrium the true force vanishes, so the residual is model error with experiment supplying the reference, from a single calculation. We previously presented this as the primary observable and that was a mistake, for a reason $H_2$ makes quantitative. Converting a residual to a bond error needs a force constant, and *which* one decides the answer: $F(R_e) = +0.1383\,E_h/a_0$ gives +27% divided by the measured $k = 0.3692$ and +53% divided by the model's own $k = 0.1875$, against a true minimum at +52%. So the conversion is right only with the model's curvature—a second quantity, itself 49% in error, requiring its own three-point scan—and the measured constant *understates* the

model's error by twenty-five points. Every $\Delta R$ in Table 3 was converted that way and is a lower bound.

What the force retains are two narrower uses, both real. The residual total force measures *electronic non-convergence*, since it equals the net force of the cloud on the nuclei and vanishes only for a stationary state. And agreement with $-dE/dR$ is what licenses using forces for dynamics at all: 0.06% in the axisymmetric reduction, 3.2% in the three-dimensional solver. Neither is a measure of how good the model is.

The two primary quantities are not independent, and reporting either alone invites a wrong generalisation: a model can have the right depth and the wrong position if its curvature is wrong, which is exactly what hydrogen does. Section 6 works both through for $H_2$, the one system whose atomic reference involves no core at all.

**What this costs, and why we pay it.** An earlier version of this work recommended the force route for its economy: one calculation per molecule, no scan, no minimum to find. That economy is real and we have given it up, because a commensurate two-sided scan is three to five calculations per molecule. The reason is in the numbers above—the cheap observable needed a second quantity to interpret, that quantity was itself 49% wrong, and using the measured one instead understated the error by twenty-five points. A validation protocol that is cheap and twenty-five points wrong is not a bargain. What survives of the original argument is the part that did not depend on cost: the model is a force law, so agreement between its force and $-dE/dR$ is a consistency requirement rather than a definition, and we return to it in Section 15.

# 3 Model and method

Each electron $m$ carries a density $u_m^2$ supported on a domain fixed by a weight $w_m$, with $\int w_m u_m^2 = n_m$ the domain's occupation. Domains compete for territory, so the partition has free internal boundaries. The energy is

$$E = \sum_m \Big[ \tfrac{1}{2} \int w_m |\nabla u_m|^2 - \int w_m K u_m^2 \Big] + E_{ee} + \sum_{a<b} \frac{Z_a Z_b}{R_{ab}}, \qquad K(\mathbf{r}) = \sum_a \frac{Z_a}{r_{\mathrm{soft},a}}, \tag{2}$$

with every term weighted by $w$, electron–electron repulsion counting each pair once by the $i \neq j$ rule, and $r_{\mathrm{soft}}^2 = |\mathbf{r} - \mathbf{R}_a|^2 + (\sigma h)^2$ a kernel softening on the grid of spacing $h$. Calculations use bare kernels ($r_c = 0$) throughout, the declared grid of Section 4, and relaxation to a fixed step count with early termination disabled—the latter because the built-in stopping criterion compares energies between wall-clock samples and was found to report convergence on a plateau from which the state was still moving.

## 3.1 How the forces are computed

The force on a nucleus is the classical electrostatic force of the charge distribution, the content of the electrostatic force theorem [4, 5]; in a model built from charge densities it is definitional rather than derived. It is obtained by differentiating the discretised energy analytically rather than by sampling its gradient. From $V_{eK} = -\sum_a Z_a \int \rho / r_{\mathrm{soft},a}$,

$$\mathbf{F}_a \;=\; -\frac{\partial V_{eK}}{\partial \mathbf{R}_a} \;=\; Z_a \int \rho(\mathbf{r})\, \frac{\mathbf{r} - \mathbf{R}_a}{r_{\mathrm{soft},a}^3}\, dV, \tag{3}$$

a volume integral over the grid. It carries no averaging radius and is never evaluated at the nuclear cusp, and because it differentiates the same softened kernel from which the energy is built it agrees with $-dE/dR$ by construction, up to the term arising from the partition boundary moving with the nuclei—the analogue here of the basis-set derivative term of Pulay [6].

The alternative of sampling $Z_a \nabla V_H$ at the nucleus and averaging over a sphere of radius $a$ to suppress grid noise is not a well-defined quantity on this mesh: holding $H_2$ fixed at $R = 1.68\,a_0$ and varying only $a$ from 0.24 to $0.96\,a_0$ carries the force from $-0.050$ to $+0.027\,E_h/a_0$, with no plateau and a change of sign, since the averaging bias is $O(a^2 \nabla\rho)$ and $\nabla\rho$ is largest exactly at the cusp. Equation (3) is used throughout; its agreement with the energy is measured in Section 6.3, where it predicts the model's own minimum to 0.2%.

# 4 One mesh for every molecule

All calculations reported here use the GPU solver `mol_fast` on a single declared grid:

$$h = 0.1\,a_0, \qquad 200^3 \text{ cells}, \qquad \text{box } 20\,a_0, \qquad \sigma = 1 \;\; (\varepsilon = h), \tag{4}$$

the same for every molecule in the list. The grid is not adjusted per system and no mesh ladder is run to justify a number. This requires defending, because it inverts the usual practice.

**At fixed $h$, the grid size is the box.** Since box $= nh$, declaring $h$ means that choosing between $100^3$, $200^3$ and $300^3$ is choosing a box of 10, 20 or $30\,a_0$—how much vacuum surrounds the molecule, not how finely it is resolved. The bond length is nearly insensitive to that: holding $h = 0.1$ and enlarging the box from 10 to $14\,a_0$ moves $H_2$ from 1.8855 to $1.8921\,a_0$, a change of 0.35%. Box 20 is chosen as the smallest that holds the polyatomic entries with room for their electron clouds.

**The box must be fixed too.** This is the reason the grid cannot follow the molecule. Comparing two systems computed in differently sized boxes inflates their energy difference: CO gave $0.0369\,E_h$ on mixed boxes against $0.0137\,E_h$ on matched ones, a factor of 2.7. A table of binding energies assembled from per-molecule boxes is therefore not a table of model errors but of box errors, and the only remedy is one box for the whole table, sized by its largest member.

**What the declaration costs.** It costs the description of the model as parameter-free. With $h$ declared, $h$ is a constant of the model, fixed by measurement and not derived from anything; and (4) is not an innocent choice, since $h = 0.1$ is the spacing at which the binding energy of hydrogen agrees with experiment while its bond is 52% too long (Section 6)—and Section 6.2 shows the bond error there to be discretisation, the converged value being short by only a few percent. A declared mesh is therefore not neutral between observables: this one is chosen where the energy is right and the geometry is not. A genuine constant of nature would be pinned by something physical; the natural candidate here is the kernel softening length $\varepsilon = h$, and at $0.1\,a_0$ that exceeds a real nuclear radius by four orders of magnitude, so no physical length currently fixes it.

**And $h$ is not the only one.** The free boundary is advanced by

$$w \;\leftarrow\; w \;+\; \lambda\,\delta t\,|c|\,\nabla^2 w \;+\; \mu\,\delta t\,c\,|\nabla w|, \qquad c \;=\; u_m - \max_{n \neq m} u_n, \tag{5}$$

clamped to $[0, 1]$, with the occupancy set to zero below a floor $w_f$ wherever another domain clearly owns the cell. The fixed point $c = 0$ is the amplitude-continuity condition of Section 6.5 and is part

of the model; the coefficients are not. Declaring the grid therefore means declaring, alongside $h$: the interface sharpness $\lambda = 0.05$, the front speed $\mu = 1$, the residue floor $w_f = 0$, the relaxation step $\delta t = \nu h^2$, the initial territory extent $R_{\text{init}} = 3\,a_0$—and the box itself, which Section 6.1 measures at four points of hydrogen's binding energy and which therefore cannot be presented as a container sized to fit the molecules. We list them because an unlisted regularisation is indistinguishable from a fitted one.

Their measured sensitivity on $H_2$ is the reason this is a short list rather than a serious one. Reducing $\lambda$ by a factor of four moves the bond by 0.4 points; doubling $R_{\text{init}}$ moves it by 0.2 points and the energy by $0.006\,E_h$. The floor is the exception: raising it from 0 to 0.05 lengthens the bond by up to 2.9 points, monotonically and saturating, and—awkwardly—the setting that leaves no spurious residue behind a retreating front makes the bond *worse*, which is why $w_f = 0$ is declared rather than defaulted quietly. The front speed $\mu$ has not been laddered on $H_2$ at all; it was fixed on a one-electron test where $\mu = 2$ was stable and $\mu = 10$ failed to conserve charge, and that is the outstanding gap in this list.

**What it buys.** Falsifiability. With $h$ free, any disagreement with experiment can be deferred to convergence, and no single calculation is ever answerable for its result. With $h$ declared, every bond length and every binding energy in the table is a prediction carrying no remaining adjustment, and the whole table stands or falls together. The sharp test, which we cannot yet run, is whether *one* spacing fits both observables at once; Section 6 shows that hydrogen's two observables currently want different ones.

## 5 The atom: homogeneous inner shells, sector-split valence

The binding energy (1) is a difference against separated atoms, so the atomic model is part of the molecular model and is stated first.

An atom is a kernel carrying the *full* nuclear charge $Z$ with every electron represented as its own charge density, arranged in radial shells. The inner shells are **homogeneous**: spherical, non-split, multi-occupancy domains, whose intra-shell repulsion the solver reduces exactly for a shared shape, retaining $(n-1)/n$ of the self-Hartree, which is the identity

$$\sum_{i<j} \iint \frac{\rho_i \rho_j}{r} \;=\; \frac{n(n-1)}{2} \cdot \frac{1}{n^2} \iint \frac{\rho\rho}{r} \;=\; \frac{n-1}{2n} \iint \frac{\rho\rho}{r}. \tag{6}$$

The outermost shell is **sector-split**: divided into angular domains rather than carried as one sphere. Oxygen is $2 + 3 + 3$ on a $+8$ kernel, the configuration obtained from atomic calculations independently of any molecular input [2].

Three properties of this construction decide what the molecular results can mean.

**The core excludes the valence by occupying space, not by decree.** There is no inner cutoff and no core radius. The valence is held out of the core region by the core electrons themselves, through the same non-overlap that separates any two domains—and that exclusion is *compressible*: a valence pressed inward can squeeze the core, where a radius would simply block it. This distinction is not cosmetic. A radius calibrated to reproduce an atom's valence energy comes out at $0.92\,a_0$ for carbon, three and a half times the $0.263\,a_0$ Slater estimate of its $1s$ radius, and two such spheres on a second-row diatomic leave only $0.50\,a_0$ of bonding region between them: the calculation then converges reproducibly to an asymmetric state. A core that yields transfers to molecules; a radius sized to fix the atom does not.

**The sector split changes the sign of the dipole, and we do not know why.** Measured on OH with everything else held: the same valence electrons give $\mu_x = +0.778$ (correct sign) as two angular domains and $-0.470$—inverted—as one. Same kernel, same electron count, same geometry; only the partition differs. The effect is large and reproducible, and it is the sharpest discriminator we have between constructions, because a sign cannot be rescued by cancellation.

We decline to explain it by saying a single domain cannot polarise, which would be wrong: a domain's density is a free field over its territory, constrained only by $\int wu^2 = n$, so it may perfectly well carry more charge on one side than the other. A domain explicitly *flagged spherical* is shape-locked and can only translate or resize, and that is the case the nested construction uses for its inner shells—but it does not apply to a free-form single domain, which is what the measurement above compares. What the split supplies is therefore not polarisability as such. The mechanism is open, and we report the measurement without it.

**Hydrogen is the one atom with nothing approximated.** One electron, one domain, no core and no shell structure, so its error is pure discretisation: $E(\mathrm{H}) = -0.4866\,E_h$ against the exact $-0.5$, too shallow by 2.7%. This is why $H_2$ is worked through first in Section 6: every error there belongs to the grid and the interface treatment, not to the atom. Helium bounds the next rung, having two electrons and still no core: it comes out 10.5% too shallow on the declared grid with its kinetic energy 24% low, because $\varepsilon = h$ cannot resolve a $Z = 2$ cusp. Deeper kernels have sharper cusps, so the softening is the dominant error in every atom heavier than hydrogen, and it is a resolution error rather than a defect of the construction.

**Scope of what is reported.** The molecular results in Section 10 were computed with the *valence-only* reduction of this model—a kernel of reduced charge $Z_x$ carrying the bonding shell alone, with the core absorbed into the charge reduction. That reduction is cheaper and it is what the tabulated numbers use, but it is not the model above: with the core electrons absent, nothing performs the exclusion, and the valence collapses to $\langle r \rangle = 0.45\,a_0$ against a physical 1.6, which over-binds carbon by a factor of 2.8 against the sum of its ionisation energies. The shell model is stated here because it is what removes that defect at its origin; its own molecular numbers are not yet measured, and nothing in Section 10 should be read as testing it.

# 6 Hydrogen

We work hydrogen through on the three observables the simulator reports: the binding energy, the geometry the model relaxes to, and the force residual at the observed geometry. $H_2$ is the natural first case because, as Section 5 notes, its atomic reference carries no pseudokernel, so every error below belongs to the discretisation and the interface treatment rather than to the core model.

Everything below is on the declared grid (4). Before the results, what the box contributes at fixed spacing, because it turns out to contribute more than a grid choice should:

| box ($a_0$) | $E(\mathrm{H})$ | $E(\mathrm{H_2})$ | $E_{\mathrm{bind}}$ | vs $D_e$ | $R$ |
|---|---|---|---|---|---|
| 10 | $-0.486810$ | $-1.147250$ | 0.173630 | $-0.5\%$ | 1.7951 |
| 12 | $-0.486509$ | $-1.145757$ | 0.172739 | $-1.0\%$ | 1.7963 |
| 14 | $-0.486591$ | $-1.143770$ | 0.170588 | $-2.2\%$ | 1.7973 |
| **20** (declared) | $\mathbf{-0.486591}$ | $\mathbf{-1.139790}$ | **0.166608** | $\mathbf{-4.5\%}$ | **1.7983** |

The atomic reference is box-free, varying by $0.0003\,E_h$ across the range, and the bond is box-free, varying by 0.18%. The whole effect is in the molecular energy, which is the interior interface again:

the molecule has one and the atom does not. The binding energy spans 4.0 points of $D_e$, twice the initialisation scatter measured below, so the dependence is real.

It cannot be extrapolated away. The shift per unit box is 0.00075, 0.00099, 0.00066 $E_h$ across the three intervals—not monotone, so not a power law in 1/box—and every individual rung is at or below the scatter even though the overall range is twice it. Neither a convergent limit nor an unbounded expansion of the free boundary is established, and we quote no box-limit value.

Two consequences follow, and both are more awkward than the grid declaration was meant to be. The box joins $h$ and the five coefficients of (5) in the declared set: it is not a convenience sized to hold the molecules but a parameter worth four points of the answer. And the dependence runs the unflattering way—the *smallest* box binds most strongly, within half a percent of $D_e$, and enlarging it to the declared value costs four points. An earlier figure of six points between boxes 12 and 15 was obtained at fixed *cell count*, so the spacing changed with the box; it is confounded and withdrawn.

The experimental targets are $R_e = 1.40028\,a_0$ and $k = 5.75\,\mathrm{mdyn/Å} = 0.3692\,E_h/a_0^2$. For the dissociation energy there are two standard values and the difference between them exceeds the model's own scatter, so we give both:

| | | $E_h$ | eV | |
|---|---|---|---|---|
| $D_0$ | measured | 0.16457 | 4.478 | the directly observed value, from the vibrational ground state |
| $D_e$ | well depth | 0.17449 | 4.748 | $D_0$ plus a zero-point energy of $0.00993\,E_h$ |

$D_e$ is the matched comparison, because the model carries no nuclear motion: what it computes is the depth of a well, not the energy of a vibrational state. But $D_e$ is a hybrid of a measurement and a spectroscopic correction, whereas $D_0$ is purely measured, and the two differ by 6.0 percentage points. We quote against $D_e$ and state the $D_0$ figure alongside it throughout rather than choose silently.

## 6.1 Binding energy

| interface kinetic treatment | $E$(H) | $E(\mathrm{H_2})$ | $E_{\mathrm{bind}}$ | vs $D_e$ | vs $D_0$ |
|---|---|---|---|---|---|
| **cut-off at $w = 0.9$, box 20** | **−0.486591** | **−1.139790** | **0.166608** | **−4.5%** | **+1.2%** |
| cut-off at $w = 0.9$, box 12 | −0.486509 | −1.145757 | 0.172739 | −1.0% | +5.0% |
| full $w$-weighting, box 12 | −0.486422 | −1.134067 | 0.161223 | −7.6% | −2.0% |
| exterior cleared, box 12 | −0.485361 | −0.935158 | −0.035564 | unbound | unbound |

Table 1: Binding energy of $H_2$ under three treatments of the partially occupied cells at the interface, with molecule and atom relaxed for the same number of steps. Note that the two conventions disagree about which of the first two rows is better, the cut-off being 1.0% low against $D_e$ and 5.0% high against $D_0$ while the uncorrected weighting is 7.6% low and 2.0% low respectively—another reason to quote both. The reference and the molecule must be run to the same step count: an earlier value of −11.4% for the middle row came from an atomic reference stopped at 70 500 steps against a molecule at 232 500, and is withdrawn.

Table 1 shows that the binding energy is decided by how the interface cells are counted. Where two domains meet, the partition weight $w$ passes through intermediate values, and the kinetic energy of those cells is an artefact of the diffuse representation of a boundary that the model treats as sharp. Excluding cells below an occupancy of 0.9 removes $0.0117\,E_h$ from the molecule and only $0.000087\,E_h$ from the atom—a ratio of 67 to 1 per domain—because the molecule has an interior

interface where two domains meet while the atom has only an outer edge with nothing on the far side. The exclusion therefore does not cancel in the difference, and it is the single change that brings the binding energy to experiment.

The third row is the control. Clearing the exterior amplitude altogether, rather than merely declining to count its kinetic energy, unbinds the molecule: $E_{\mathrm{bind}}$ goes negative and the relaxed separation runs out to $3.56\,a_0$. The amplitude just outside a domain is what makes the interface flux small, and removing it imposes a drain at the boundary instead.

**The uncertainty is measured, not estimated.** The exterior amplitude in `mol_fast` is not varied by the relaxation—where $w$ vanishes the update is identically zero—so its value is inherited from the initial state, and the binding energy could in principle be an artefact of that choice. Varying the initial decay constant by a factor of 2.1 gives

| $z_{\mathrm{init}}$ | 0.7 | 1.0 | 1.5 |
|---|---|---|---|
| $E_{\mathrm{bind}}$ | 0.175248 | 0.172739 | 0.176339 |
| vs $D_e$ | +0.45% | −0.99% | +1.07% |
| $R$ (relaxed) | 1.7993 | 1.7963 | 1.7953 |

a spread of 2.1 percentage points which is not monotone in the seed, so it is scatter rather than a dependence, and a geometry spread of 0.22%. We therefore quote

$$E_{\mathrm{bind}}(\mathrm{H_2}) \;=\; 0.1666{\pm}0.0018\,E_h \;=\; -4.5{\pm}1.0\%\ \text{of}\ D_e \;=\; +1.2{\pm}1.1\%\ \text{of}\ D_0 \qquad \text{[declared box; +4.0 points at box} \tag{7}$$

the central value on the declared grid, the stated uncertainty the measured initialisation scatter, and the bracketed figure the declared-box sensitivity—which is twice the scatter and must be quoted with it. The result lies *between* the two experimental conventions, $D_0 = 0.16456 < 0.16661 < D_e = 0.17447$, closer to the directly measured $D_0$ than to the well depth it should properly be compared with. So the model's binding is accurate to a few percent on either reading, and which few percent depends on two things it cannot itself resolve: a convention, and a declared box. For scale, the choice of interface condition discussed in Section 6.5 moves the same quantity by 29 to 58 points, so the initialisation is a minor term.

## 6.2 Relaxed geometry

Releasing the kernels under the force and letting the system find its own minimum gives the geometry the model actually predicts. For $\mathrm{H_2}$ at the declared grid this is

$$R_{\mathrm{min}} = 2.134\,a_0 \qquad (+52.4\%\ \text{against}\ R_e = 1.40028), \tag{8}$$

taken from a fixed-geometry scan on commensurate separations $D = (2i+1)h$, which avoids the spurious corrugation that appears when kernels are placed at arbitrary sub-cell offsets. The force zero of the same scan lies at $2.066\,a_0$, agreeing with the energy minimum to 3.2%.

A one-sided relaxation released from the observed geometry stops at $1.796\,a_0$ (+28.3%), well short of the minimum, and that is a systematic bias rather than a better answer: released from inside, the molecule expands and the relaxation terminates while the force is still outward. The two-sided bracket is 1.85 to $2.09\,a_0$. We report +52%, which is the model's own answer, and note that any relaxation started at the experimental geometry flatters the model by roughly ten points.

**Refining the mesh does not converge on experiment.** The one-sided relaxed bond over three spacings is

| $h$ | 0.1395 | 0.1000 | 0.0800 |
|---|---|---|---|
| $R$ | 1.8897 | 1.7963 | 1.6999 |
| $dR/dh$ | — | 2.365 | 4.820 |

with the local slope *steepening* rather than flattening. Extrapolating linearly in $h$ from the two finest points gives $R(h \to 0) = 1.314\,a_0$, which is 6% *below* experiment, and crosses the measured 1.40028 at $h \approx 0.018$.

That crossing is not convergence on the right answer, and we can now say so from a converged calculation rather than an extrapolation. In three dimensions $h = 0.02$ is out of reach—$500^3$ cells at the smallest legitimate box—but $H_2$ is exactly a two-domain system on an axis, which an axisymmetric reduction treats exactly. Scanning commensurate separations $D = (2m + 1)h$ there gives

| $h$ | 0.050 | 0.040 | 0.030 | 0.025 | 0.020 |
|---|---|---|---|---|---|
| $D_{\min}$ | 1.3707 | 1.3682 | 1.3686 | 1.3687 | 1.3682 |
| force zero | 1.3699 | 1.3702 | 1.3691 | 1.3684 | 1.3685 |

a bond flat to 0.18% across a 2.5-fold refinement and already converged by $h = 0.04$. The minimum and the force zero agree to 0.06% at every spacing, which is the sharpest confirmation available that the force is the gradient of the energy. The converged bond is $1.368\,a_0$**, or** $-2.3\%$: short, and not for want of resolution.

So both interface conditions undershoot—$-2.3\%$ with the strict condition, $-6\%$ extrapolated with the weak one—and neither reaches experiment at any spacing. The $+52\%$ of (8) is therefore almost entirely *discretisation*, and the construction's own bond error is a few percent.

**Why the coarse-grid minimum sits long.** The repulsive side of the balance is exact: $V_{KK} = 1/R$ is arithmetic on the geometry built. So the equilibrium is wherever the electronic attraction gives out, and at $h = 0.1$ it gives out early. Against the exact values at $R_e$, the model's kinetic energy is 17% low (0.9724 against 1.1745) and its electronic potential 13% weak ($-2.6683$ against $-3.0631$)—both signatures of a density too diffuse, because $\varepsilon = h$ cannot resolve the cusp. The kernels then profit by separating, since their repulsion is exact and falls as $1/R$: moving from 1.400 to $2.134\,a_0$ sheds $0.2455\,E_h$ of kernel repulsion that was never earned by screening.

A 13% weakening alone would move the minimum only some 6%, since $A'(R) = 1/R^2$ at the minimum and $R^2$ scales inversely with $A'$. The remaining factor is the curvature, 49% too soft, which converts a small imbalance into a large displacement. One cause—the unresolved cusp—with two multiplying consequences: the weak attraction sets the direction and the flat well sets the size. That is also why the depth survives while the position does not: on a well that flat, sitting 52% long costs $\frac{1}{2}k\Delta R^2 = 0.051\,E_h$. The crossing at $h \approx 0.018$ is a positive discretisation error passing through zero as it cancels a negative construction error; declaring that spacing would fit one number to one observable and predict nothing. The caveat belongs on the axisymmetric figure too: it carries a different softening coefficient, but $\varepsilon = \sigma h$ vanishes with $h$ either way, so both schemes approach the same bare-Coulomb problem and the interface condition is the only difference that survives the limit.

### 6.3 Forces at the observed geometry

The check of Section 6.3 needs no reference calculation and no minimum-finding, and that cheapness is what originally recommended it. At the measured geometry the true force on every nucleus is zero, so the computed residual is model error read directly, with experiment supplying the reference. For $H_2$ at $R_e$,

$$F(R_e) = +0.1383\, E_h/a_0 \qquad \text{(outward: the bond wants to lengthen)}, \tag{9}$$

one calculation rather than a scan. The residual converts to a displacement through $\Delta R \approx F/k$, and *which* $k$ appears there matters:

$$\frac{F}{k_{\text{model}}} = \frac{0.1383}{0.1875} = 0.738\, a_0 \quad \Longrightarrow \quad R = 2.138\, a_0, \tag{10}$$

against the scanned minimum of 2.134—agreement to 0.17%. The force at the observed geometry predicts the model's own equilibrium, which establishes that the volume integral of Section 3.1 is the gradient of the energy the solver reports, and it does so from a single calculation.

**A correction to the conversion.** Using the *measured* force constant instead gives $F/k = 0.375\, a_0$, or +26.8%, which understates the model's bond error by the stiffness ratio $k/k_{\text{model}} = 1.97$. The measured $k$ answers a different question—what bond error this force residual would produce in a well of the correct stiffness—and it is the model's own curvature that governs where the model's minimum lies. Bond errors converted with the measured $k$ are therefore systematically too small whenever the model's well is too soft, and we quote $\Delta R$ from the model's own curvature below, at the cost of a three-point scan per molecule in place of one calculation.

### 6.4 One defect, not three

The three observables are not three independent verdicts. Collecting them against the measured curvature $k = 0.3692\, E_h/a_0^2$:

| | | |
|---|---|---|
| depth | $E_{\text{bind}} = 0.1666 \pm 0.0018$ | $-4.5 \pm 1.0\%$ |
| curvature | $k_{\text{model}} = 0.1875$ | $-49\%$ |
| position | $R_{\text{min}} = 2.134$ | $+52\%$ |

the depth is right, the gradient is consistent with the energy to 0.2%, and the position is wrong *because* the well is too soft: a curvature 49% low turns a given force residual into a displacement nearly twice as large. The error is one quantity, the curvature, with the bond length as its consequence.

One qualification, from the converged calculation of Section 6.2: the soft well is a property of the *weak interface treatment*, not of the model. The same system under the strict condition gives curvatures scattering about $0.41\, E_h/a_0^2$, some 11% *above* experiment rather than 49% below. So "one defect, the curvature" is a statement about this solver's front, and the two interface conditions bracket the true curvature from opposite sides exactly as they bracket the bond. The reason the position can be so wrong while the depth is right is that the well is flat: sitting 52% long costs $\frac{1}{2}k\Delta R^2 = 0.051\, E_h$ on the model's own curve, and any quantity differenced near the bottom of a well that shallow is insensitive to where along it the minimum sits.

### 6.5 The interface condition, and why hydrogen cannot decide it

The free boundary between two domains needs a condition, and the construction admits two that are both defensible. The model requires $\Psi \in H^1$, that is continuity of the densities across the interface, whose natural condition is continuity of the amplitude, $u_1 = u_2$, with the normal derivative free to jump. `mol_fast` implements this: its front advances on the local amplitude mismatch and converges where that mismatch vanishes. The alternative treats the domains as disjoint with $u$ unconstrained at its own boundary, for which the natural condition is $\partial u/\partial n = 0$; a mirrored ghost value, or equivalently omitting the face from the stencil, imposes it exactly, and two independent implementations in this programme do so.

The two give materially different energies—+27.9% and +58.0% on $E_{\mathrm{bind}}$ for the strict condition against +0.2% for the continuity condition, with bond errors of +15.9% and −2.3% against +52%—so the choice is not a detail. But $H_2$ cannot adjudicate it. A symmetric diatomic has $u_1 = u_2$ at the midplane *and*, by reflection, $\partial u/\partial n = 0$; the two conditions coincide, and the spread above is a difference between discretisations rather than between physical conditions. Settling it requires a two-domain system with no symmetry plane, which is not in the present list.

## 7 The solver

The measurements above were made with a GPU solver (`mol_fast`) that carries one domain per entry and caps at sixteen domains. That cap is the binding constraint on molecular size, and its meaning depends on the construction: with one domain per atom carrying $Z_x$ electrons it permits sixteen atoms, which covers benzene, glycine, ethanol and comparable small organics; with the one-electron domains of the underlying construction it permits sixteen electrons, which covers $CH_4$ and $C_2H_2$ but not benzene.

The scope of the pseudo-kernel core model itself, measured on two-centre systems, is reported separately [3].

Two properties of `mol_fast` are relevant to reading the numbers above, and we state them rather than leave them implicit. The partition weights are diffuse: $w$ passes through intermediate values over a few cells at an interface instead of switching between 0 and 1, which is what makes the interface kinetic energy of Section 6.1 a quantity that must be decided rather than a consequence of the model. And the amplitude outside a domain is not relaxed—where $w$ vanishes the update is identically zero—so it is inherited from the initial state. The measured consequence of that is the ±1.0% band on the binding energy, which is small, and bounds how much of the result can be attributed to initialisation rather than to the model.

A second implementation (`molecule.js`) carries the same model as one global field partitioned by labels, enforcing true non-overlap with sharp interfaces, and admits two thousand domains; it has been run on alpha-helices and on chignolin. It is not used for any number reported here. Its interest is that its sharp partition removes the interface ambiguity above at the cost of a different force treatment, so it is the natural setting for the asymmetric two-domain test that Section 6.5 identifies as outstanding.

## 8 Cost

A volume integral over the grid is more expensive than a point gradient, and it is worth stating by how much. The point gradient visits $N_{\mathrm{atoms}} \times \frac{4}{3}\pi R^3 \approx 270\,N$ cells for a four-cell sphere. The volume form visits $N_{\mathrm{atoms}} \times S^3$, about $10^6 N$ on a $101^3$ grid: some four thousand times more.

The relevant comparison, however, is with what the solver already does. One volume-force evaluation costs about the same as one update sweep of the electronic field, and *less* than one evaluation of the potential, which requires an elliptic solve of several sweeps per electron. Forces are moreover not evaluated every step: once per animation frame in one implementation, every ten electronic steps in the other, giving overheads of well under one percent and of order ten percent of the update cost respectively—the latter adjustable, since nuclei move slowly compared with the electronic relaxation.

At protein scale the $N \times S^3$ product would warrant a cutoff radius. It is worth noting that such a cutoff behaves oppositely to the smoothing radius it replaces: truncating a convergent integral has an error that *falls* with radius, and distant neutral groups contribute as dipole/$r^3$ rather than charge/$r^2$, so a plateau in the cutoff is attainable and meaningful. The smoothing radius of the point gradient admitted no plateau because its bias *grew* as $a^2$. The same word denotes a convergent truncation in one case and a divergent bias in the other.

## 9 Three checks, and which one applies

Every calculation carries three internal checks. They are not interchangeable, and the choice between them is dictated by symmetry—a point we found easy to get wrong.

**Kernel–kernel energy.** $V_{KK} = \sum_{a<b} Z_a Z_b / R_{ab}$ is exact arithmetic and confirms the geometry *actually built*, independently of any physics. Every configuration reported below matched its predicted value; this is how fractional kernel placement, per-bond spacing and coincident kernels were verified.

**Total force as a convergence criterion.** The sum of nuclear forces is *not* automatically zero. It equals the net force the electron cloud exerts on the nuclei; only when the electronic state is stationary does the cloud exert no net force, and then $\sum_a \mathbf{F}_a = 0$ follows by Newton's third law. The residual is therefore a direct measure of electronic non-convergence—and a more reliable one than any energy-based criterion, since a relaxation can sit on a long plateau with the energy changing imperceptibly while the state is far from settled.

**But symmetry cancels it.** In a symmetric molecule, mirror or rotational symmetry forces $\sum_a \mathbf{F}_a = 0$ regardless of accuracy. A small residual in $CO_2$ (we measure 0.06% of the largest force) establishes only that the algorithm preserves symmetry. The sum carries information only in directions symmetry does not protect: every direction for a heteronuclear molecule, the $C_3$ axis of $NH_3$, the $C_2$ axis of water.

**Per-atom zeros.** Where symmetry is complete, the useful check is instead a single atom's force. An atom at a centre of symmetry—the carbon of $CO_2$ or of $CH_4$—must feel exactly nothing. Unlike the sum, this cannot be satisfied by cancellation among atoms, so a nonzero value is unambiguously error and its magnitude is the per-atom floor.

# 10 Results: relaxed geometries and binding energies, with forces as a check

Table 2 is the paper's primary table: the two predicted quantities with the dipole beside them. Table 3, which follows, is the force *check* and the older binding-energy set, kept because it is the only coverage of the polyatomic rows.

| system | $Z_x$ | $E_{\mathrm{bind}}$ | $\mu$ (D) | $\mu_{\mathrm{exp}}$ | $\lvert\sum F\rvert$ | state |
|---|---|---|---|---|---|---|
| $N_2$ | 3,3 | 0.2131 | 0.038 | 0 | 1.2% | accepted |
| $O_2$ | 3,3 | 0.2156 | 0.012 | 0 | 5.4% | accepted |
| CO | 4,3 | 0.2219 | 0.154 | 0.110 | 6.3% | accepted |
| $CO_2$ | 4,3,3 | 0.3440 | 0.000 | 0 | 0.1% | accepted |
| $N_2O$ | 3,3,3 | 0.3915 | 0.023 | 0.161 | 7.4% | accepted |
| $CH_4$ | 4,1,1,1,1 | 1.0823 | 0.004 | 0 | 3.4% | accepted |
| HCN | 1,4,3 | *(0.4371)* | *(0.165)* | 2.985 | 83% | rejected |
| $H_2O$ | 3,1,1 | *(0.5164)* | *(0.938)* | 1.855 | 117% | rejected |
| $NH_3$ | 3,1,1,1 | *(0.6834)* | *(0.786)* | 1.472 | 127% | rejected |

Table 2: Primary results, one grid throughout (box $16\,a_0$, $h = 0.10$, $r_c = 0$, occupancy cut-off 0.9, 232 500 steps) against atomic references measured identically: $E(\mathrm{H}) = -0.486591$, $E(\mathrm{C}) = -15.113441$, $E(\mathrm{N}) = E(\mathrm{O}) = -7.252538$. **The state column is an acceptance test.** Where symmetry requires $\mu = 0$ ($N_2$, $O_2$, $CO_2$, $CH_4$) the computed value is a pure error measurement needing no reference, and all four pass—0.000 to 0.038 D. Where no symmetry protects the total force (CO, $N_2O$, HCN) the residual is a genuine convergence measurement and admits rows below 20%. Rejected rows are shown parenthesised so the rejection can be checked, never as results; three of the four have an *unprotected direction*—$H_2O$ and $NH_3$ a $C_2$ or $C_3$ axis, HCN no symmetry at all—and settle to a *stable* asymmetric state rather than converging slowly: water's residual held at 116.7, 116.5, 116.6% across its run with its two hydrogens equivalent to 0.1%, so the imbalance lies along the axis and more steps will not remove it. **Hydrogen is deliberately absent.** Its repeat run on this grid did not converge, while it converges at boxes 12 and 20—so that is a bad grid point, not a failed molecule, and a different thing entirely from the rejections above, which are *stable* wrong states. Listing it here would conflate the two. Hydrogen is treated in full in Section 6, where its binding is −4.5% of $D_e$, its bond converges to $1.368\,a_0$, and its force agrees with $-dE/dR$ to 0.06%. $R_{\mathrm{min}}$ is omitted: it exists for $H_2$ alone, no commensurate scan having been run for the others. The two polar accepted rows give +40% (CO) and −86% ($N_2O$) against measurement—CO's dipole being famously small at 0.11 D—so the dipole magnitudes are poor where the null tests are excellent.

The ten remaining rows are being measured on this grid. Until they are, the binding energies in Table 3 stand as the only polyatomic coverage, and seven of its nine populated entries come from the superseded grid—annotated there, but old.

Table 3 collects the configurations. Forces are quoted with + meaning outward, so a positive entry indicates the bond wants to lengthen. Where a configuration fails its own check we report the failure rather than a bond error.

Two columns of this table require the qualifications established on hydrogen, and we give them before the table rather than after it. The $\Delta R$ entries were converted with the *measured* force constant, so by Section 6.3 each is a *lower bound* on the model's bond error, too small by the ratio of the measured curvature to the model's own; for $H_2$ that ratio is 1.97 and the entry below

should read +52% rather than +30%. Correcting the remaining rows requires a three-point scan per configuration to measure each model curvature, which has not been done. And the last column is a diagnostic, not a score: an $H_2$ configuration that does not bind at all returns 2.1% on it, which would place it among the best entries here. Its only legitimate reading is the symmetry-dependent one given in the caption.

**Only the declared valences appear.** Every row is at $Z_x = 1, 4, 3, 3$. The C+2 and O+2 variants are not results of the model but the alternatives it rejects, measured as such in Sections 12 and 13, and carrying them here would present a ruled-out convention as a finding. Removing them has one consequence worth stating plainly: **CO and $CO_2$ leave the table altogether**, because the only force calculations ever made for them were at O+2. There is no force measurement for either molecule at the valences the simulator actually uses, and the earlier reading of $CO_2$'s $-7.2\%$ as an ordinary bounded error was therefore a statement about a configuration this paper rejects. What is measured for $CO_2$ at $3, 4, 3$ is its binding energy, and it does not bind (Section 11).

**Precision of the binding-energy column.** The six decimals are the solver's, not the measurement's, and most of this column carries no established precision at all. Its error is dominated by a *systematic*: the one row measured on both grids moves by $0.0555\,E_h$, or 32 points of $D_e$ ($H_2$: 0.1111 against 0.1666), which is larger than several of the entries themselves—$N_2$ at 0.0155, CO at 0.0137, $O_2$ at 0.0770. Mixing boxes between rows contributes a further $0.023\,E_h$ (CO 0.0369 against 0.0137 matched), the box at fixed $h$ some 0.007, and the step count a similar amount, these rows having been stopped at 119 500 steps against the 232 500 used elsewhere. No *random* component was ever measured on them: no seed variation and no box ladder. The operational consequence is a single rule—**an entry below about $0.056\,E_h$ cannot be distinguished from zero on this evidence**—and $N_2$ is the demonstration, its $-0.0155$ becoming $+0.2764$ once the grid was matched. Only two rows carry a stated uncertainty: $H_2$ at $0.1666 \pm 0.0018$ from measured initialisation scatter, with a further four points of declared-box sensitivity; and $N_2$ at $+0.2764$ with a 6% reference-shift uncertainty, which secures the sign and not the value.

The binding-energy column carries a third and sharper qualification. Only hydrogen is on the declared grid; the rest predate it, and the single row available on both grids shifts by 35 points of $D_e$ between them. That shift is larger than every deficit in the column, so no entry here establishes that a molecule fails to bind—a statement we had drawn from this set and now withdraw. What the column does establish is the direction and rough scale of the construction's binding on a common grid, and that the earlier grid understated it severely.

### 10.1 $H_4$ against an exact reference

One configuration can be scored against an exact method rather than experiment. Full configuration interaction in a minimal basis [9] places the equilibrium of the equally spaced $H_4$ chain at $a = 1.8\,a_0$, and the chain has one electron per site with no pseudopotential, so no core parameter is involved. At that geometry all four RealQM forces point outward—the model would expand a chain the exact method calls stationary—while over-binding by 14–18% per atom when each method is referred to its own atom. Too long is the pattern $H_2$ shows at +52% (Section 6.2). The over-binding is less clear-cut than it reads: the reference is a full configuration-interaction calculation in a minimal STO-6G basis, whose cohesive energy is itself 14% low against reality (0.0766 against $0.0872\,E_h$ per atom), so over-binding against it is not the same statement as over-binding against experiment.

## 11 Binding energies across compositions

Hydrogen's binding energy is accurate because of a cancellation, not because its energies are accurate: the total is 2.4% too shallow and 94% of that error cancels against the atomic reference, leaving the $0.0017\,E_h$ residual that is the binding error. The cancellation is that complete because molecule and reference share a composition and, for $H_2$, because the reference is a one-electron domain with no pseudokernel at all. Neither holds for the rest of the list, and the arithmetic of how badly it fails is worth setting out, because it decides which binding energies can be believed.

The pseudokernel atoms are over-bound by the factors of Section 5: against the sum of their ionisation energies, carbon by 2.57 and the $Z_x = 3$ oxygen by 2.03. Converting those to absolute terms, each carbon reference is too deep by $8.55\,E_h$ and each oxygen by $3.47\,E_h$, and the same error enters the molecule.

*Provisional figures.* The two factors were measured on an earlier grid (box 15, $h = 0.1389$, $\varepsilon = 0.14$). Re-run on the declared grid (4) and converged, carbon is $-15.106\,E_h$ rather than $-13.999$, a factor of 2.78 and an absolute error of $9.67\,E_h$. The finer grid is *deeper*, so the over-binding is worse where this paper's calculations actually live, and the numbers in the table below are therefore lower bounds. The oxygen repeat is in progress. The conclusion drawn from the table is unaffected in direction—a deeper reference makes the required cancellation more stringent, not less—but no figure here should be quoted until both are on one grid.

What matters is the error's size relative to the bond energy being extracted:

| system | common error carried in | $E_{\mathrm{bind}}$ | ratio |
|---|---|---|---|
| $H_2$ | $0.03\,E_h$ | $+0.1666$ | **0.2** |
| $CH_4$ | $8.61\,E_h$ | $+0.4081$ | 21 |
| $CO_2$ | $15.50\,E_h$ | $-0.2110$ | **73** |

*This argument has since been refuted, and by its own test.* It rested on $N_2$ and $CO_2$ failing to bind, requiring cancellations of 99.8% and 98.6% against the atomic over-binding. Re-measured with molecule and reference on one grid, **both bind**: $N_2$ by $+0.2131$ and $CO_2$ by $+0.3440\,E_h$, against the recorded $-0.0155$ and $-0.211$. The $CO_2$ swing alone is $0.555\,E_h$. So the premise was not the atom model at all—it was differencing across grids, twice, and the ratios computed from it were circular besides, their denominators being the model's own near-zero bindings.

What survives is weaker and still worth saying: the atomic references *are* over-bound, by a factor of 2.8 for carbon, and that error does enter every binding energy and does grow with composition. What does not survive is any claim that it prevents binding. On one grid the six accepted rows of Table 2 all bind, including the two that were the headline failures.

For $H_2$ the common error is a sixth of the signal, so *no* cancellation is needed to get the sign right—the molecule would bind even if the two errors did not cancel at all—and the 94% that does cancel is what turns a sound sign into a 1% magnitude. For $CO_2$ the common error is seventy-three times the signal, so the cancellation must be complete to 98.6% merely to return the right *sign*. It is not, and $CO_2$ comes out unbound by $0.211\,E_h$. Nothing is specifically wrong with $CO_2$. It is the composition in this list where the atom model's over-binding is largest relative to what the binding energy is trying to measure.

This gives the existing rule—differences within one composition are trustworthy, differences across compositions are not—a quantitative form, and it says where the remedy has to be applied. No improvement to the interface treatment or the grid can rescue $CO_2$'s binding energy, because the defect is in the atomic reference and enters at fifteen Hartree. The pseudokernel atom model is the thing that has to improve—but Section 5 does *not* identify what in it is wrong, and we are careful not to imply otherwise. The two candidates there are testable and untested.

## 12 Kernel depth decides convergence

The clearest results hold geometry, symmetry, mesh and every other input fixed and change only the central kernel's charge:

| | shallow | deep | factor |
|---|---|---|---|
| $CH_4$ (identical $T_d$ geometry) | C+2: 40% | C+4: 0.6% | 60 |
| HCN (identical geometry) | C+2: 88.7% | C+4: 5.5% | 16 |

A deep kernel holds its domain and fixes its boundary against the neighbours; a shallow one leaves the partition undetermined. The direction is worth emphasising: C+4 is the *less* resolved kernel, at $Zh = 0.48$ against a working guideline of $Zh \lesssim 0.1$, so depth helps convergence more than coarse resolution hurts it.

Three alternative explanations were tested and excluded.

- **Not hydrogen content.** $CH_4$ carries four hydrogens and is the cleanest polyatomic measured; HCN carries one and reaches 5.5%; CO at 4,2 carries none and fails at 38.5%. Hydrogen is neither necessary nor sufficient for failure.
- **Not the box.** The net force on water is 0.216, 0.215, 0.214 in boxes of 12, 18 and 24 $a_0$ with the mesh held fixed—0.9% across a doubling of the box. The state is box-converged and the residual is a property of it. The control calculation on $H_2$ changes *sign* with box and with mesh, marking its own error as stochastic.
- **Not off-axis placement.** Tetrahedral ligands are maximally off-axis and $CH_4$ is nearly exact, because the cube diagonals are a symmetry of the grid.

### 12.1 Cubic-grid anisotropy

A second, smaller effect is visible in planar $NH_3$, which has $D_{3h}$ symmetry and must therefore give $\sum_a \mathbf{F}_a = 0$ in every direction; it gives 70% of the largest force along $x$. Resolving by ligand: the one lying on the grid axis has radial force 0.0802, the two off-axis ones 0.0545 and 0.0544. The off-axis pair agrees with itself to 0.2% and differs from the on-axis ligand by 47%.

A cubic grid has fourfold symmetry in its planes and threefold symmetry about its body diagonals. Molecular geometries that share those symmetries—linear, and tetrahedral—are represented evenly; threefold-in-plane and bent geometries are not. This accounts for $NH_3$ and $H_2O$ being the two worst configurations measured and $CH_4$ among the best, and it is a property of the discretisation rather than of the model.

## 13 Valence conventions, measured

$Z_x$ is a modelling choice, and the protocol makes it testable. A depth ladder at fixed mesh, scored both by the symmetry-breaking error and by the bond error:

| | O+2 | O+3 | O+6 (single domain) | N+3 | N+5 |
|---|---|---|---|---|---|
| pair asymmetry | 19.5% | **3.7**% | 1.4% | 6.0% | 6.1% |
| $\Delta R$ | −6.4% | **−7.1**% | +59% | −8.1% | −13.7% |

O+3 is the better oxygen: a fivefold gain in convergence over O+2 at the same bond error. N+3 stands, full valence costing accuracy while buying no convergence. Most instructive is O+6 as a single six-electron domain, which converges best of all and yields a bond 59% too long—it converges accurately to the wrong answer, which is exactly the failure mode the protocol is designed to make visible. So a rule of "use the deepest $Z_x$" is false beyond the valence structure the atomic model selects; depth helps only where the domain was genuinely under-bound.

It is worth noting that O+3 is one of the two valence shells in the atomic configuration 2+3+3 obtained independently from atomic calculations [2], a convergence of atomic and molecular evidence on the same number. Carbon admits no corresponding reduction: C+2 would freeze into the core the $2s$ pair that hybridises into the bonds, cannot form four bonds, and—since $Z_x$ is the only label distinguishing elements—makes carbon identical to O+2.

# 14 Kernel dynamics

Releasing the kernels and letting them move under the force is both the sharpest test of the force and the capability a simulator is finally for. It is also where the distinction between a force and a sampled gradient stops being academic: a force that is not the gradient of anything conserves no energy, so in sustained dynamics it pumps or drains the system indefinitely.

Released from either side of the expected minimum, $H_2$ relaxes to

| start | relaxes to | residual force | direction |
|---|---|---|---|
| 1.40 | 1.8502 | $[-0.0178, +0.0057]$ | expanded |
| 2.40 | 2.0932 | $[+0.0128, -0.0168]$ | contracted |

Both move the correct way and bracket $[1.850, 2.093]$, whose midpoint 1.972 sits 0.5% from the energy minimum at 1.963 and inside the force's own zero crossing of 1.84–1.89 measured at fixed geometry. Three independent routes—force zero, energy minimum, and dynamical relaxation from both directions—therefore agree to a few percent, where before the force was repaired the first two disagreed by 27% and the force pointed the wrong way at $R = 1.68$.

The bracket is not a failure to converge but a measured precision. The residual energy drift between runs is $\sim 0.011\,E_h$; through the curvature $k = 0.370\,E_h/a_0^2$ that corresponds to a displacement $\sqrt{2(0.011)/0.370} = 0.24\,a_0$, and the bracket is $0.243\,a_0$ wide. The two relaxations converged to exactly the precision the unconverged electronic state permits, and tightening the geometry requires tightening the electronics, not the dynamics.

Two cautions follow, both measured rather than anticipated. The relaxed states carry energies about $0.011\,E_h$ *above* the fixed-geometry scan at comparable separation, the electronic field lagging behind moving nuclei in the familiar way; so relaxation yields trustworthy geometry and untrustworthy energies, and the two should be taken from different calculations. And at three centres the electronic relaxation does not settle at all, oscillating by $\sim 0.03\,E_h$ out to $3 \times 10^5$ steps, so kernel dynamics there would track a wandering density. Dynamics is at present a two-centre capability.

## 14.1 What the old force does in dynamics

The same relaxation under the sphere-averaged gradient settles at 1.5203 with residual forces $[+0.01821, +0.00020]$—both of the same sign, so the molecule carries a net translational force and drifts bodily, which no isolated system may do. In sustained dynamics that is a spurious acceleration of the whole system, and in a thermostatted simulation it would appear as heating indistinguishable from physical dissipation.

It also lands *closer to experiment*: 1.520 against the measured 1.401, where the repaired force gives 1.97. This is worth stating plainly because it inverts the naive test. The repaired force agrees with the model's own energy minimum to 0.5%; it reports faithfully what the model says, and what the model says is wrong by 40%. The old force was wrong *about the model* in a way that partly compensated for the model's own error. Validating a force against experiment alone would have preferred the defective one; only the internal checks—the lone atom feeling exactly zero, the vanishing of $\sum_a \mathbf{F}_a$, and agreement with $-dE/dR$—reject it. Fixing the instrument made agreement with measurement worse and moved the error from the instrument to the model, which is where it can be addressed.

## 15 Relation to $-dE/dR$

In a force-law model the force is definitional and the geometry is where it vanishes, so agreement with the energy gradient is not required for the model to be well posed. It remains worth checking for three reasons. A force that is not a gradient conserves no energy, which in dynamics produces spurious heating indistinguishable from physical dissipation. It is a strong and free check on implementation: it is what exposed the sign-flipping force of Section 3.1, which was otherwise unremarkable—smooth, antisymmetric, and crossing zero at a plausible separation. And once the implementation is correct, the residual disagreement measures the contribution from the partition boundary moving with the nuclei, a term specific to a free-boundary model and not otherwise accessible.

## 16 Bearing on previously recorded three-centre failures

Earlier work in this programme recorded that every three-centre system attempted descended monotonically inward through its experimental bond, across nuclear charges, with and without a core, for odd and closed-shell systems alike. Measured by force at the observed geometry instead, $CO_2$ gives $-7.2\%$ and HCN $-7.8\%$: ordinary bounded errors of the same size as the diatomics, not a collapse. Those scans were assembled from an energy relaxation that we find oscillates by $\sim 0.03\,E_h$ out to $3 \times 10^5$ steps, read through a stopping criterion that compares energies between wall-clock samples and cannot detect it. The recorded failures are therefore consistent with artefacts of the scan rather than properties of the model, and the systems should be re-measured before any mechanism is proposed for them.

## 17 Limitations

The bond errors reported here are not competitive with established electronic structure methods, and are not offered as such; they measure where this real-space construction currently stands. Several specific limitations bound the numbers.

The construction is not parameter-free. Besides the declared spacing it carries the five coefficients of the free-boundary update listed in Section 4, four of which have measured sensitivities below three points on $H_2$ and one of which ($\mu$, the front speed) has never been laddered on a molecule. It also takes the valence charges $Z_x$ from separate atomic work, and the kinetic occupancy cut-off at 0.9 is a declared choice worth 6.6 points of binding energy. None of these is fitted to the quantities reported here, but none is derived either.

All calculations use bare kernels, so the deepest, $Z = 4$, sits at $Zh = 0.48$, far above the resolution guideline. Mesh ladders are deliberately *not* used to justify the numbers, for the reason

given in Section 4; the one ladder we do report, for $H_2$'s bond, shows the refinement sequence passing through experiment rather than converging on it, which is a limitation of the construction and not of the sampling.

The convergence problem is characterised but not solved. We can now measure electronic non-convergence, attribute it principally to shallow kernels, and exclude three rival explanations; we cannot yet remove it. Configurations containing shallow kernels or threefold and bent geometries remain unusable, which includes water and ammonia—a serious limitation for any application to hydrogen-bonded systems.

Measured geometries, force constants and dissociation energies are taken from the standard compilations [7, 8] and are recorded as unverified in the source files. Dissociation energies carry the further ambiguity of Section 6.1: $D_0$ is measured and $D_e$ adds a spectroscopic zero-point correction, the two differ by six points for $H_2$, and only $D_e$ is the quantity a model without nuclear motion computes.

Finally, the construction pins the electron count of each domain, so charge transfer between unlike atoms is representable only through the shape of each domain and not through its occupation. For homonuclear systems this costs nothing; for a heteronuclear molecule it constrains precisely the asymmetry that defines it.

## 18 Conclusion

The simulator is defined by a declared grid rather than by a convergence limit: one spacing for every molecule, with the atomic references that enter the binding energy becoming constants of the model rather than per-run bookkeeping. That choice gives up the description of the construction as parameter-free, and we have listed what it actually carries. What it gains is that each number is answerable: with the grid fixed there is no remaining adjustment to defer a disagreement to.

Hydrogen, the one system whose atomic reference involves no pseudokernel, comes out as a single defect rather than three. Its depth is right to a few percent—$0.1666 \pm 0.0018\,E_h$, falling between the two experimental conventions, with the uncertainty measured over a factor of two in the initialisation rather than estimated. Its gradient is right, the volume force predicting the model's own minimum to 0.2% from one calculation at the observed geometry. Its curvature is 49% too soft, and its bond is 52% too long *because* of that, through $\Delta R = F/k_{\mathrm{model}}$. One quantity is wrong and the other errors follow from it, which is a more useful finding than three independent discrepancies would have been, and it identifies the curvature as what any improvement has to address.

**And nitrogen binds.** Measured with molecule and reference on one grid, $E_{\mathrm{bind}} = +0.2764\,E_h$ in a symmetric state, where the same quantity computed across two grids gave $-0.0155$ and was reported as a failure to bind the strongest common bond. That reversal is the single most consequential result here, and it is a lesson about method rather than about nitrogen: a number measured on one grid is not evidence about another. The same comparison moves hydrogen's binding by thirty-two points and carbon's energy by $1.1\,E_h$, and it is why the binding column of Table 3 carries a systematic larger than several of its own entries—nothing below about $0.056\,E_h$ in it can be distinguished from zero.

Three further things bear on how the earlier protocol should be read. The residual total force is a convergence diagnostic and not a figure of merit: an unbound $H_2$ scores as well on it as the best entries in the table. Bond errors converted with the measured force constant are lower bounds, too small by the stiffness ratio wherever the model's well is soft—twenty-five points for $H_2$. And the long bond is a *resolution* artefact, not a property of the construction: an axisymmetric reduction,

exact for two domains on an axis, converges the same bond to $1.368\,a_0$ and holds it flat to 0.18% under a two-and-a-half-fold refinement, so the model's own bond error is $-2.3\%$ where the declared grid reports $+52\%$.

What remains outstanding is specific rather than general. The front-speed coefficient has never been laddered on a molecule. The choice of interface condition, worth 29 to 58 points of binding energy, cannot be settled on any symmetric system—$H_2$ included—because symmetry makes the candidates coincide. The atomic references are the binding energies' real bottleneck, the valence-only reduction over-binding carbon by a factor of 2.8; the shell construction of Section 5 is stated as what removes that at its origin, and its molecular numbers are not yet measured. And the two conventions for a dissociation energy differ by more than the model's own scatter, so every binding figure here is quoted against both.

## References


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[2] C. Johnson, *RealQM for Atoms and the Periodic Table*. The shell configurations used here, including oxygen as 2+3+3, are obtained there from atomic calculations independent of any molecular input.

[3] C. Johnson, *The Two-Centre Test: where the pseudo-kernel core model works, and where it stops*. Companion study measuring three core models—hard depletion, clamp and empty core—against the alkali and halogen dimers.

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[9] Full configuration interaction in a minimal (STO-6G) basis for equally spaced hydrogen chains $H_N$, generated for this work: equilibrium at $a = 1.8\,a_0$ with $E = -2.19038$ ($N = 4$) and $-3.26674$ ($N = 6$), each referred to its own atom $E(\mathrm{H}) = -0.471039$. The hydrogen chain is a standard benchmark for many-electron methods, having one electron per site and no pseudopotential, so no core parameter enters the comparison.

[10] Public code, the molecule simulator of Section 1, and an interactive gallery of all experiments in the present work: `https://claes542.github.io/RealMolecule/gallery.html` (source repository: `https://github.com/Claes542/RealMolecule`).

| system | $Z_x$ | force | $\Delta R$ from $F/k_{\mathrm{exp}}$ ∗ | $\lvert\sum F\rvert/\lvert F\rvert_{\max}$ | $E_{\mathrm{bind}}$ § |
|---|---|---|---|---|---|
| $H_2$ | 1,1 | +0.1383 | +30% ∗ | 14.7% † | **+0.1666** |
| $N_2$ | 3,3 | −0.2505 | −8.2% | 3.7% † | **+0.2764 ¶** |
| $O_2$ | 3,3 | −0.2463 | −14.3% | 8.4% † | +0.0770 |
| **HCN** | **1,4,3** | C≡N −7.8%, H–C +3.5% | | **5.5**% | +0.6099 |
| $H_2O$ | 3,1,1 | — | | 126% ‡ | +0.4307 |
| $H_3$ lin. | 1,1,1 | +0.0420 | — | 34.9% † | — |
| $H_4$ chain | 1,1,1,1 | +0.0316 | — | 7.4–16% † | — |
| $C_2H_2$ | 1,4,4,1 | outward | — | 25% † | +0.3177 |
| $C_2N_2$ | 3,4,4,3 | N inward, C outward | | 3.2% † | +1.9819 |
| $NH_3$ | 3,1,1,1 | — | | 141% ‡ | +0.5424 |
| $NH_3$ planar | 3,1,1,1 | — | | 70% ‡ | — |
| $CH_4$ | 4,1,1,1,1 | 0.036 outward | — | 0.6% † | +0.4081 |

Table 3: The force check at measured geometries, with binding energies. **Neither of the paper's two primary observables appears in full here**: the model's own relaxed geometry is measured only for $H_2$ (Section 6.2, +52% from a commensurate scan), and the $\Delta R$ column is *not* that quantity—it is the force residual divided by the *experimental* force constant, which understates the model's error by twenty-five points on the one row where both are known. The $|\sum F|$ column is a convergence diagnostic to which no verdict attaches, because what it *measures* depends on symmetry. †: every direction is symmetry-cancelled, so the entry reports whether the algorithm preserves symmetry and bounds nothing about convergence. ‡: cancelled except along the unique axis, so the residual is a convergence measurement in that direction. Unmarked rows (CO, HCN) have no symmetry, and their entries are genuine convergence measurements—note that these are among the largest in the table. Rows where no $\Delta R$ is given are those whose own checks exceed the signal. ∗: converted with the *experimental* force constant, hence a lower bound on the model's bond error. For $H_2$ the measured minimum is +52% against this column's +30% (Section 6.2), and the same correction applies in unknown measure to every other row, none of which has had its geometry scanned. §: only the $H_2$ entry (bold) is on the declared grid of (4) with the $w = 0.9$ cut-off; every other binding energy comes from an earlier set at the default valences in a box of $15\,a_0$ with $h = 0.1389$, $\varepsilon = 0.14$ and the uncorrected kinetic accounting. ¶: $H_2$ is the one row measured both ways, and it moves by $+0.0555\,E_h$, or 32 points of $D_e$, between them—6.6 of those points from the cut-off and the rest from the grid and softening. Every deficit in this column is smaller than that shift, so *the unbound entries are not results.* That has now been measured rather than suspected: re-run with molecule and reference on one grid (box 10, $h = 0.10$), $\mathbf{N_2}$ **binds by** $+0.2764\,E_h$, from $E(\mathrm{N}) = -7.260914$ and $E(\mathrm{N_2}) = -14.798207$, with a symmetric state ($\Sigma F = 5.5\%$). The atomic reference shifts by only $0.008\,E_h$ between boxes 10 and 16, i.e. 6% of the binding energy, so the sign is secure. The earlier $-0.0155$ came from box 15 with $h = 0.1389$ and is withdrawn. Blank entries are rows whose valences or geometry were not in the earlier set; a row is deliberately left empty rather than filled from a run at different $Z_x$. There is no experimental column because the repository holds no verified table of atomisation energies, and $H_2$ is the only row with a reference we have checked. Configurations in which two different elements receive the same $Z_x$ are degenerate—the model cannot distinguish them—and are omitted.